

Ayush Maurya

ayushmaurya2003@gmail.com | ayush.maurya@research.iiit.ac.in | +91-7985149173 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Builds research prototypes that ship. Published an open-source agent-based electoral simulation as a PyPI package (ElectoralSim), then translated that simulation expertise into a co-founded startup (CrowdTwin) for predictive crowd dynamics. Bridges research questions and production systems: designing controlled ML experiments, deploying LLM pipelines, and communicating findings without jargon. Precision over volume.

EDUCATION

IIIT Hyderabad	2021–Present
B.Tech in Computer Science and M.S. by Research in Computing and Human Sciences	
CHD: Computer Science and Humanities Integrated Dual Degree (Offered by the Human Sciences Research Center)	
Mary Gardiner's Convent School	2021
Class XII — 90.1%	
City Montessori School	2019
Class X — 94.2%	

EXPERIENCE

SERC Digital Twin	Environmental IoT
- Developed a real-time environmental monitoring dashboard using the MERN stack. - Integrated local sensors for temperature, humidity, water, and air quality monitoring.	
TAFEA	System Design
- Designed a student client software system using standard SDLC methodologies. - Integrated LLM APIs to support tailored student-facing features and workflows.	

PROJECTS

ElectoralSim	Agent-Based Electoral Simulation
- Published open-source research prototype for vectorized agent-based electoral simulation on PyPI. - Implemented modular simulation framework supporting multiple electoral systems with documented API.	
ML for Local Geoguesser	Visual Geolocation via Deep Learning
- Designed controlled experiments comparing ResNet50 architectures for visual geolocation inference. - Achieved 92% region accuracy through iterative hypothesis testing and statistical model evaluation.	
Highland History Project	OCR + LLM Pipeline
- Built OCR-LLM pipeline digitizing historical documents for Himalayan trade-network research. - Evaluated multiple LLM configurations for degraded-text recovery with accuracy benchmarks.	
Design for Social Innovation (TAFEA)	AI-Assisted Education Platform
- Developed AI-assisted platform enabling curriculum design for resource-constrained classrooms. - Conducted human-centered research and evaluated prototype effectiveness through user studies.	
Age Prediction Model	Transfer Learning vs Ensemble
- Conducted comparative study of transfer learning versus ensemble methods for age prediction. - Formulated experimental hypotheses, implemented in PyTorch, and documented statistical findings.	

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL, HTML/CSS
Technologies: Machine Learning, PyTorch, Linux, System Design, Systems Programming
UI/UX and Design: Wireframing, Prototyping, User Stories, User Research, Interaction Design

HONOURS

UGEE Rank: 111 out of 40,000+ candidates

ACTIVITIES

Bass guitarist in college band and lawn tennis player.